

is similar to that of seeding clouds with a chemical to form rain. The idea is that only seeds available in the chamber are those of ions produced by the interaction with radiation. The resulting trail of alcohol droplets can be seen against the black background in the bottom of the chamber.

These are only a few of the devices commonly utilized for purposes of detection and measurement of radioactivity and radiation. It is important to understand that when working with radioactivity/radiation, due to our inability to sense radiation, we need them to assist us in detecting the presence of radiation and we also need them to help monitor the radiation.

OTHER DETECTION DEVICES

Another common device used for detection and measurement is the ionization chamber. The Geiger counter, survey meter, and personal dosimeters work on the basis of the ionization chamber. The principle operation of an ionization chamber is that it will produce an electric current in the presence of a radioactive source. Ionization chambers consist of tubes filled with gas, such as argon. When radiation enters the tube and interacts with the gas, it removes electrons from the gas. The gas atoms become positively charged ions, and the free electrons move through the gas to a wire in the tube, setting up a current. The current is commonly amplified and sent to a recording or counting device. This in response may produce a flash of light, ticking sounds, or an analog readout. Ionization chambers are capable of measuring the amount of radiation by means of measuring the amount of current produced.



Measurement of Radioactivity

For measuring radioactivity, three types of devices are available:

1. Gas-filled tube counters e.g. the Geiger Muller Counter
 2. Scintillation Counters
 3. Semi-conductor Detectors
1. **The Geiger Counter:** A potential difference just below that required to produce a discharge is applied to the tube (1000 V). Any atoms of the gas struck by the γ -rays entering the tube are ionized, causing a discharge. Discharges are monitored and counted by electronic circuitry and the output is reported as counts/sec or Rontgens/hr or mR/hr.
 2. **Scintillation Counters:** Crystals of certain substances e.g. cesium fluoride, cadmium tungstate, anthracene and sodium iodide emit small flashes of light when bombarded by γ -rays. The most commonly used phosphor in scintillation counters is NaI with a minute quantity of thallium added. In the instrument, the crystal is positioned against a photocell which in turn is linked to a recording unit. The number of flashes produced per unit time is proportional to the intensity of radiation. Small portable scintillation counters are available.